

C. AUTOREGRESSIVE MODEL DESIGN

The use of autoregressive models for excitation force prediction is based on the assumption that the excitation force estimates $\hat{f}_{ex,k}$ at time k are a linear combination of past estimations as follows:

$$\hat{f}_{ex,k} = a_{1,k}\hat{f}_{ex,k-1} + a_{2,k}\hat{f}_{ex,k-2} + \dots + a_{p,k}\hat{f}_{ex,k-p} \quad (58)$$

where p indicates the order of the AR model, whereas its coefficients are time varying to handle sea state variations, and expressed at time k by $a_{i,k}$, $\forall i = 1, 2, \dots, p$. The above equation can be formulated in matrix form as follows:

$$\hat{f}_{ex,k} = A_k^T \hat{F}_{ex,k} \quad (59)$$

where $A_k = [a_{1,k} \ a_{2,k} \ \dots \ a_{p,k}]^T \in \mathbb{R}^{p \times 1}$ is the coefficient vector and $F_{ex,k} = [\hat{f}_{ex,k-1} \ \hat{f}_{ex,k-2} \ \dots \ \hat{f}_{ex,k-p}]^T \in \mathbb{R}^{p \times 1}$ is the vector of past excitation force estimations. The problem of determining the coefficients can be transformed into a least-squares problem with the objective of minimizing the one-step-ahead prediction error, given by

$$\min_{A_k^*} E \left[\left(\hat{f}_{ex,k} - A_k^T \hat{F}_{ex,k} \right) \left(\hat{f}_{ex,k} - A_k^T \hat{F}_{ex,k} \right)^T \right] \quad (60)$$

In [35], it was found that A_k^* can be calculated iteratively using a recursive least-squares algorithm that can update the coefficients with each new excitation wave estimate. The algorithm is summarized as follows:

$$A_k = A_{k-1} + \theta_k \left[\hat{f}_{ex,k} - A_{k-1}^T \hat{F}_{ex,k} \right] \quad (61)$$

$$\theta_k = \frac{P_{k-1} \hat{F}_{ex,k}}{r + \hat{F}_{ex,k}^T P_{k-1} \hat{F}_{ex,k}} \quad (62)$$

$$P_k = \frac{1}{\psi} \left[P_{k-1} - \frac{P_{k-1} \hat{F}_{ex,k} \hat{F}_{ex,k}^T P_{k-1}}{\psi + \hat{F}_{ex,k}^T P_{k-1} \hat{F}_{ex,k}} \right] \quad (63)$$

with $\theta_k \in \mathbb{R}^{p \times 1}$ being the least squares weighting factor, and $P_k \in \mathbb{R}^{p \times 1}$ is the matrix that is proportional to the previous estimate variance. The notation ψ denotes the forgetting factor that has a value of $0 \ll \psi < 1$. The initial values of A_0 and P_0 were set to $A_0 = 0_{p \times 1}$ and $P_0 = cI_{p \times p}$, where c is a very large scale. Once the coefficients are obtained, it is possible to obtain l step-ahead predictions as follows:

$$\hat{f}_{ex,k+l|k} = \sum_{i=1}^p a_{i,k} \hat{f}_{ex,k+l-1|k}. \quad (64)$$

The above equation shows that past excitation force estimations for the one-step-ahead prediction $\hat{f}_{ex,k+1|k}$ are actual past estimations from the SR-UKF. For the two-step-ahead prediction $\hat{f}_{ex,k+2|k}$, one of the past data points is the newly predicted $\hat{f}_{ex,k+1|k}$ from the previous step. The same is observed for consecutive predictions; some of their past data are predictions from the previous steps. More accurate excitation force predictions can be obtained by considering multi-step-ahead error minimization instead of the single step in (60). However, no analytical solution A_k^* can be calculated because of the nonlinear nature of the

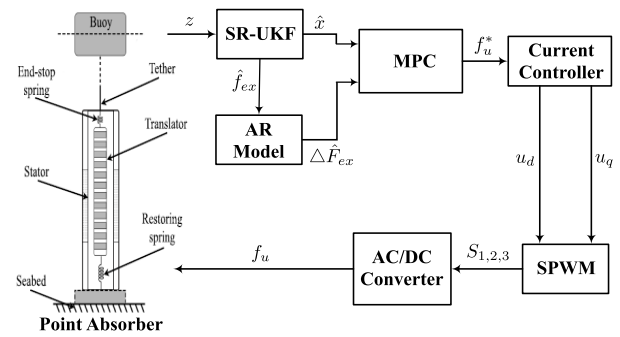


FIGURE 2. The proposed MPC framework.

multi-step-ahead error minimization problems. Some studies have proposed Gaussian Newton algorithms [36] or extended Kalman filtering [37] to find local A_k^* values. The accuracy of the excitation force predictions is not a priority in this work; therefore, only one-step-ahead error minimization is considered to avoid computational complexities in multi-step-ahead error minimization problems.

IV. MODEL PREDICTIVE CONTROL

In this section, a model predictive control (MPC) framework, as shown in Fig. 2, is developed to improve the energy capturing efficiency of a point absorber type WEC. MPC requires knowledge of the system states, in addition to current and future estimates of the wave excitation force received from the SR-UKF and AR models. The MPC predicts the future trajectory of the point absorber over the prediction horizon N_p . These trajectory predictions are then used in the MPC cost function, which focuses on maximizing the wave energy capture while including the system constraints and physical limitations. Quadratic programming (QP) is used to optimize the cost function over N_p , and a sequence of control forces are generated. The MPC then sends the first optimal control force in sequence as a q -axis current setpoint to the PI current controller. The control of the PMLG stator currents allows for the manipulation of forces produced by the PMLG, and hence realizes the optimal control force required for maximum energy extraction.

A. PREDICTION MODEL

The predicted future trajectory requires the discretization of the linear state-space model in (27) and (28). The first-order hold is used for discretization as follows:

$$x_{d,k+1} = A_d x_{d,k} + B_d \Delta f_{u,k+1} + F_d \Delta f_{ex,k+1} \quad (65)$$

$$y_{d,k} = C_d x_{d,k} \quad (66)$$

where

$$\Delta f_{u,k+1} = f_{u,k+1} - f_{u,k}, \quad \Delta f_{ex,k+1} = f_{ex,k+1} - f_{ex,k}$$

$$\begin{aligned}
x_{d,k} &= \begin{bmatrix} z_k \\ \dot{z}_k \\ x_{r,k} \\ f_{u,k} \\ f_{ex,k} \end{bmatrix} \in \mathbb{R}^{8 \times 1}, \quad y_{d,k} = \begin{bmatrix} z_k \\ \dot{z}_k \\ f_{u,k} \end{bmatrix} \in \mathbb{R}^{3 \times 1} \\
A_d &= \begin{bmatrix} e^{A_c T_{mpc}} & \Lambda & \Lambda \\ 0_{1 \times 4} & 1 & 0 \\ 0_{1 \times 4} & 0 & 1 \end{bmatrix} \in \mathbb{R}^{8 \times 8}, \\
C_d &= \begin{bmatrix} 1 & 0 & 0_{1 \times 6} \\ 0 & 1 & 0_{1 \times 5} \\ 0_{1 \times 5} & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 8}, \\
B_d &= \begin{bmatrix} \Gamma \\ 1 \\ 0 \end{bmatrix} \in \mathbb{R}^{8 \times 1}, \quad F_d = \begin{bmatrix} \Gamma \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^{8 \times 1}, \\
\Lambda &= A_c^{-1} \left(e^{A_c T_{mpc}} - I_{1 \times 6} \right) B_c \in \mathbb{R}^{6 \times 1}, \\
\Gamma &= \frac{1}{T_{mpc}} = A_c^{-1} (\Lambda - T_{mpc} B_c) \in \mathbb{R}^{6 \times 1}. \quad (67)
\end{aligned}$$

The subscript d indicates the discretized quantities and the sampling time T_{mpc} corresponds to the MPC sampling rate. Assuming that a prediction horizon N_p is selected, the output prediction vector $y_{d,k}$ can be obtained over N_p as in [12]

$$\hat{y}_{d,k} = P x_{d,k} + W_B \hat{\Delta F}_{u,k} + W_F \hat{\Delta F}_{ex,k} \quad (68)$$

where

$$\begin{aligned}
\hat{y}_{d,k} &= \begin{bmatrix} y_{d,k+1|k} \\ y_{d,k+2|k} \\ \vdots \\ y_{d,k+N_p|k} \end{bmatrix}, \quad P = \begin{bmatrix} C_d A_d \\ C_d A_d^2 \\ \vdots \\ C_d A_d^{N_p} \end{bmatrix}, \\
W_B &= \begin{bmatrix} C_d B_d & 0 & \dots & 0 \\ C_d A_d B_d & C_d B_d & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_d A_d^{N_p-1} B_d & C_d A_d^{N_p-2} B_d & \dots & C_d B_d \end{bmatrix}, \\
W_F &= \begin{bmatrix} C_d F_d & 0 & \dots & 0 \\ C_d A_d F_d & C_d F_d & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_d A_d^{N_p-1} F_d & C_d A_d^{N_p-2} F_d & \dots & C_d F_d \end{bmatrix}, \\
\hat{\Delta F}_{u,k} &= \begin{bmatrix} \Delta f_{u,k+1|k} \\ \Delta f_{u,k+2|k} \\ \vdots \\ \Delta f_{u,k+N_p|k} \end{bmatrix}, \quad \hat{\Delta F}_{ex,k} = \begin{bmatrix} \Delta f_{ex,k+1|k} \\ \Delta f_{ex,k+2|k} \\ \vdots \\ \Delta f_{ex,k+N_p|k} \end{bmatrix}. \\
\hat{y}_{d,k} &\in \mathbb{R}^{3N_p \times 1}, \quad P \in \mathbb{R}^{3N_p \times 8}, \\
W_B &\in \mathbb{R}^{3N_p \times N_p}, \quad W_F \in \mathbb{R}^{3N_p \times N_p}. \quad (69)
\end{aligned}$$

Both $\hat{\Delta F}_{u,k}$ and $\hat{\Delta F}_{ex,k}$ are N_p dimensional column vectors. The excitation force predictions $\hat{\Delta F}_{ex,k}$ are obtained from the AR model by setting the step-ahead prediction $l = N_p$, whereas the state estimates $x_{d,k}$ are obtained from the SR-UKF.

B. COST FUNCTION

The cost function is related to the MPC objective, which is to maximize the amount of wave energy absorbed by the point absorber over the entire prediction horizon, as expressed by

$$J = - \int_t^{t+T_h} (f_u(t) \cdot \dot{z}(t) d\tau) \quad (70)$$

Here, T_h denotes the time interval of the prediction horizon, and a negative sign indicates that energy is absorbed by the waves. A discrete form of (70) can be obtained by applying the trapezoidal rule to the indefinite integral as follows:

$$\begin{aligned}
J_k &= \sum_{i=k+1}^{k+N_p-1} f_{u,i|k} \dot{z}_{i|k} + \frac{1}{2} f_{u,k+N_p|k} \dot{z}_{k+N_p|k} \\
&\quad + \sum_{i=k+1}^{k+N_p} \lambda |f_{u,i|k}|^2 \quad (71)
\end{aligned}$$

The discretized cost function in (70) differs from (71) in that it is converted to a minimization problem instead of a maximization problem by omitting the negative sign. Furthermore, a penalty term based on a non-negative weighting factor λ is included to govern MPC aggressiveness [12], [38]. The weighting factor governs aggressiveness by controlling the slew rate of the control action rather than imposing constraints on the incremental variation of the control action, which leads to higher computational burdens. However, attention should be paid when determining its value because greater values result in a large loss of absorbed energy. The findings of [38] showed that its effect on energy absorption is negligible when $\lambda \gg 1$. It is worth mentioning that (71) diverges from conventional MPC cost functions in the absence of the least-squares problem, which minimizes the tracking errors between the setpoint commands and actual plant performance. Nevertheless, the cost function can be formulated as a quadratic problem, expressed in matrix form, to include the output prediction vector $\hat{y}_{d,k}$ as follows:

$$J_k = \hat{y}_{d,k}^T Q \hat{y}_{d,k} + \hat{\Delta F}_{u,k}^T (\lambda I_{N_p \times N_p}) \hat{\Delta F}_{u,k} \quad (72)$$

where

$$\begin{aligned}
Q &= \begin{bmatrix} R_W & 0 & \dots & 0 \\ 0 & R_W & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{2} R_W \end{bmatrix} \in \mathbb{R}^{3N_p \times 3N_p} \quad (73) \\
R_W &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}
\end{aligned}$$

substituting (68) into (72) and neglecting the constant terms yields

$$\begin{aligned}
J_k &= \frac{1}{2} \hat{\Delta F}_{u,k}^T \left(W_B^T Q W_B + \lambda I_{N_p \times N_p} \right) \hat{\Delta F}_{u,k} \\
&\quad + \hat{\Delta F}_{u,k}^T W_B^T Q \left(P x_{d,k} + W_F \hat{\Delta F}_{ex,k} \right) \quad (74)
\end{aligned}$$

The optimal control sequence $\Delta \hat{F}_{u,k}$ is computed by solving (74) as a QP problem provided that it is a convex function [39]. The weighting factor λ guarantees convexity when is set to be larger than the minimal eigenvalue of $W_B^T Q W_B$, thus ensuring that $(W_B^T Q W_B + \lambda I_{N_p \times N_p})$ is positively definite. The $\Delta \hat{F}_{u,k}$ is updated online at each time step T_{mpc} , based on recent state estimates and excitation force predictions. Unfortunately, the online calculation of $\hat{F}_{u,k}$ for real-time applications is significantly affected by its dimensions [40]. The control sequence consists of N_p decision variables that must be optimized in a time interval equal to or less than T_{mpc} . A larger number of decision variables increases the computational burden to a point where solutions are unavailable during T_{mpc} . To decrease the number of decision variables, this study adopted discrete Laguerre functions to approximate $\hat{F}_{u,k}$ based on [15] and [40]. The Laguerre function is expressed in the following vector form:

$$L_{k+1} = A_L L_K \quad (75)$$

where

$$A_L = \begin{bmatrix} \alpha & 0 & \cdots & \cdots & 0 \\ \beta & \alpha & \ddots & \ddots & 0 \\ -\alpha\beta & \beta & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ (-\alpha)^{N_L-2}\beta & (-\alpha)^{N_L-3}\beta & \cdots & \beta & \alpha \end{bmatrix}$$

$$L_K = \begin{bmatrix} l_{1,k} \\ l_{2,k} \\ l_{3,k} \\ \vdots \\ l_{N_L,k} \end{bmatrix}, \quad \text{as } L_0 = \sqrt{\beta} \begin{bmatrix} 1 \\ -\alpha \\ \alpha^2 \\ \vdots \\ (-1)^{N_L-1} \alpha^{N_L-1} \end{bmatrix}$$

$$A_L \in \mathbb{R}^{N_L \times N_L}, \quad L_k \in \mathbb{R}^{N_L \times 1}. \quad (76)$$

A_L , l_k , and N_L represent the discrete state matrix of the model, Laguerre function at time k , and number of Laguerre functions, respectively. The symbol α is the scaling factor of the Laguerre function, ranging between $0 \leq \alpha \leq 1$, and is used to calculate $\beta = (1 - \alpha^2)$. The control sequence $\hat{F}_{u,k}$ can be approximated as follows:

$$\Delta \hat{F}_{u,k} = \Psi \sigma_k \quad (77)$$

where

$$\Psi = [L_1 \ L_2 \ L_3 \ \cdots \ L_{N_p}]^T \in \mathbb{R}^{N_p \times N_L}$$

$$\sigma_k = [\eta_1 \ \eta_2 \ \eta_3 \ \cdots \ \eta_{N_L}]^T \in \mathbb{R}^{N_L \times 1} \quad (78)$$

Here, $(\eta_1, \eta_2, \dots, \eta_{N_L})$ are called Laguerre coefficients, and are updated every T_{mpc} through solving the online optimization problem. Unlike the Laguerre coefficients, ψ is calculated only once as it depends only on N_L and α . Proper selections of N_L and α play a vital role in the approximation of $\hat{F}_{u,k}$. Larger values of N_L provide better approximations, yet $\hat{F}_{u,k}$ can be well approximated by a value $N_L \geq N_p$.

Moreover, a larger α can model a long prediction horizon even with a smaller N_L . Finally, substituting (77) into (78) yields

$$J_k = \frac{1}{2} \sigma_k^T \Omega \sigma_k + \sigma_k^T H \quad (79)$$

where

$$\Omega = \Psi^T (W_B^T Q W_B + \lambda I_{N_p \times N_p}) \Psi$$

$$H = \Psi^T W_B^T Q (P x_{d,k} + W_F \Delta \hat{F}_{ex,k}) \quad (80)$$

Similar to (74), the QP problem shown in (79) is also convex because proper tuning of λ can ensure $\Omega \succ 0$. Consequently, the unconstrained optimum solution σ^* in (79) is given as

$$\sigma_k^* = -H \Omega^{-1}. \quad (81)$$

From (79), it is clear that using Laguerre functions reduces the number of decision variables that must be optimized from $\hat{F}_{u,k} \in \mathbb{R}^{N_p \times 1}$ to $\sigma_k \in \mathbb{R}^{N_L \times 1}$. This reduction lowers the computational burden of the QP problem, and hence increases its feasibility for real-time applications.

C. CONSTRAINTS

The major advantage of the MPC is its ability to impose constraints on the QP problem in (79) to ensure safe point-absorber operations within their physical limitations. The constraints on the control force magnitude, heave displacement, and heave velocity can be expressed as

$$F_{min} \leq f_{u,k} \leq F_{max} \quad (82)$$

$$Z_{min} \leq z_{u,k} \leq Z_{max} \quad (83)$$

$$\dot{Z}_{min} \leq \dot{z}_{u,k} \leq \dot{Z}_{max} \quad (84)$$

where *min* and *max* denote the minimum and maximum limits, respectively. Because this MPC is based on Laguerre functions, (82)-(84) are written in terms of σ as

$$F_{min} \Theta \leq \Phi \Psi \sigma_k + f_{u,k-1} \Theta \leq F_{max} \Theta \quad (85)$$

$$Z_{min} \Theta \leq U_z (P x_{d,k} + W_b \Psi \sigma_k + W_F \Delta \hat{F}_{ex,k}) \leq Z_{max} \Theta \quad (86)$$

$$\dot{Z}_{min} \Theta \leq \dot{U}_z (P x_{d,k} + W_b \Psi \sigma_k + W_F \Delta \hat{F}_{ex,k}) \leq \dot{Z}_{max} \Theta \quad (87)$$

where

$$\Phi = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}, \quad \Theta = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix},$$

$$U_z = \begin{bmatrix} q_z & 0 & \cdots & 0 \\ 0 & q_z & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q_z \end{bmatrix}, \quad q_z = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T,$$

$$U_z = \begin{bmatrix} q_z & 0 & \cdots & 0 \\ 0 & q_z & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q_z \end{bmatrix}, \quad q_z = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}^T, \quad (88)$$

$$\Phi \in \mathbb{R}^{N_p \times N_p}, \quad \Theta \in \mathbb{R}^{N_p \times 1},$$

$$U_z \in \mathbb{R}^{N_p \times 3N_p}, \quad U_{\dot{z}} \in \mathbb{R}^{N_p \times 3N_p}.$$

The linear inequality constraints in (85), (86), and (87) can be incorporated into (79) to form the standard constrained QP problem as follows:

$$\min_{\sigma_k} \frac{1}{2} \sigma_k^T \Omega \sigma_k + \sigma_k^T H$$

$$\text{subject to} \quad \begin{bmatrix} -\Phi \Psi \\ \Phi \Psi \\ -U_z W_B \Psi \\ U_z W_B \Psi \\ -U_{\dot{z}} W_B \Psi \\ U_{\dot{z}} W_B \Psi \end{bmatrix} \sigma_k$$

$$\leq \begin{bmatrix} -F_{\min} \Theta + f_{u,k-1} \Theta \\ F_{\max} \Theta - f_{u,k-1} \Theta \\ -Z_{\min} \Theta + U_z (P x_{d,k} + W_F \Delta \hat{F}_{ex,k}) \\ Z_{\max} \Theta - U_z (P x_{d,k} + W_F \Delta \hat{F}_{ex,k}) \\ -\dot{Z}_{\min} \Theta + U_{\dot{z}} (P x_{d,k} + W_F \Delta \hat{F}_{ex,k}) \\ \dot{Z}_{\min} \Theta - U_{\dot{z}} (P x_{d,k} + W_F \Delta \hat{F}_{ex,k}) \end{bmatrix} \quad (89)$$

At each T_{mpc} , an active set algorithm is used to solve the constrained QP problem in (90). Further details regarding the active set algorithm adopted are provided in [40]. The computed Laguerre coefficients are then substituted into (77) to obtain the control sequence solution $\Delta \hat{F}_{u,k}$. Note that only the first control variable $\Delta \hat{F}_{u,k+1|k}$ is considered and added to the previous control force to obtain the recent optimal control force. In the next time step, the state estimates and excitation force predictions are updated, and the QP problem is repeated.

D. THE MPC STABILITY ANALYSIS

The stability criteria of linear time-invariant systems are inapplicable to unconstrained MPCs because the constraints convert the QP problem into a nonlinear control problem. However, it is possible to prove the unconstrained MPC stability by adding a terminal state constraint [40].

Lemma 1: The closed loop MPC is asymptotically stable by assuming that at each time instant k , the cost function (72) can be optimally minimized by solving for σ_k^* under inequality constraints and an additional terminal state constraint in the receding horizon cost function problem, which is $x_{d,k+N_p|k} = 0$.

Proof: Since the cost function (72) is guaranteed to be positive definite because of λ , it is selected as a suitable Lyapunov function candidate at time k , expressed as

$$v_k = \hat{y}_{d,k}^T Q \hat{y}_{d,k} + \Delta \hat{F}_{u,k}^T (\lambda I_{N_p \times N_p}) \Delta \hat{F}_{u,k}. \quad (90)$$

Here, $\Delta \hat{F}_{u,k} = \psi_{\sigma_k} = [L_1 \sigma_k \ L_2 \sigma_k \ \dots \ L_{N_p} \sigma_k]^T$, where the vector σ_k is the solution of (69). At time $k+1$, the Lyapunov function is given by

$$V_{k+1} = \hat{y}_{d,k+1}^T Q \hat{y}_{d,k+1} + \Delta \hat{F}_{u,k+1}^T (\lambda I_{N_p \times N_p}) \Delta \hat{F}_{u,k+1} \quad (91)$$

where $\Delta \hat{F}_{u,k+1} = [L_1 \sigma_{k+1} \ L_2 \sigma_{k+1} \ \dots \ L_{N_p} \sigma_{k+1}]^T$. The next step is to establish a relationship between the Lyapunov functions at times k and $k+1$. Although it is impossible to obtain σ_{k+1} optimally at time instant k , a feasible solution is used by shifting the elements of $\Delta \hat{F}_{u,k}$ to the left and replacing the last element with zero.

$$\hat{V}_{k+1} = \hat{y}_{d,k+1}^T Q \hat{y}_{d,k+1} + \Delta \hat{F}_{u,k+1}^T (\lambda I_{N_p \times N_p}) \Delta \hat{F}_{u,k+1} \quad (92)$$

where $\Delta \hat{F}_{u,k+1} = [L_2 \sigma_k \ L_{N_p} \sigma_k \ \dots \ 0]^T$. The use of a feasible solution implies that the cost function (72) is not optimally minimized ($V_{k+1} \leq \hat{V}_{k+1}$); thus, the difference between v_k and V_{k+1} is bounded.

$$V_{k+1} - V_k \leq \tilde{V}_{k+1} - V_k. \quad (93)$$

Because V_k and $\tilde{V}_{k+1}$ share most of the control sequences ($L_2 \sigma_k, L_3 \sigma_k, \dots, L_{N_p} \sigma_k$), their difference is given by

$$\tilde{V}_{k+1} - V_k = \frac{1}{2} y_{d,k+N_p|k+1}^T R W y_{d,k+N_p|k+1} - y_{d,k+N_p|k+1}^T R W y_{d,k+N_p|k+1} - \lambda \Delta f_{u,k+1|k}^2. \quad (94)$$

Imposing the equality terminal state constraint on (94) yields

$$\tilde{V}_{k+1} - V_k = -y_{d,k+1|k}^T R W y_{d,k+N_p|k+1} - \lambda \Delta f_{u,k+1|k}^2. \quad (95)$$

As λ is a non-negative value, $\tilde{V}_{k+1} - V_k \leq 0$, and hence $V_{k+1} - V_k \leq 0$, which implies that the closed loop MPC is asymptotically stable.

E. CURRENT CONTROL

The objective of current control is to realize the MPC control force by manipulating the PMLG stator currents through an AC/DC converter. The converter control signals are augmented into the PMLG model in (18) and (19), as follows:

$$\frac{di_d(t)}{dt} = \frac{-R_s i_d(t)}{L_s} + \omega_e(t) i_q(t) - \frac{v_{dc}(t) u_d(t)}{L_s} \quad (96)$$

$$\frac{di_q(t)}{dt} = \frac{-R_s i_q(t)}{L_s} + \omega_e(t) i_d(t) - \frac{\omega_e(t)}{L_s} \varphi_{pm} - \frac{v_{dc}(t) u_q(t)}{L_s} \quad (97)$$

where $u_d(t)$ and $u_q(t)$ represent the dq components of the control signal for the converter switches, and $v_{dc}(t)$ denotes the DC-link voltage. To regulate stator currents $i_d(t)$ and $i_q(t)$,

TABLE 1. Point absorber parameters.

Parameter (symbol)	Value
Buoy diameter (D)	5 m
Total mass (m)	3×10^4 kg
Buoy added mass at infinite freq. (m_∞)	2.85×10^4 kg
Buoy surface area (A_w)	19.6 m ²
Buoyancy stiffness coef. (C_b)	1.9×10^2 N/m
Restoring spring stiffness coef. (C_{rs})	6×10^4 N/m
Sea water density (ρ)	1.025×10^3 kg/m ³
Gravitational acceleration (g)	9.8 m/s ²
PMLG flux linkage (φ_{pm})	19.8 Wb
PMLG pole pitch (τ_p)	4.5×10^{-2} m
Number of turns per-phase (N_p)	550
PMLG stator resistance (R_s)	2 Ω
PMLG stator inductance (L_s)	81 mH
DC bus capacitor (C_{dc})	42.7 mF
End spring stiffness coefficient (C_{es})	5×10^5 N/m
Stribeck velocity threshold (v_s)	1 m/s
Normal force (F_{cn})	1×10^4 N
Static friction coefficient (μ_s)	1
Dynamic friction coefficient (μ_d)	1
Viscous friction coefficient (μ_v)	1
Max. allowable heave displacement (μ_l)	3 m

two PI control loops were used to generate the converter control signals.

$$u_d(t) = K_{dp} (i_d^*(t) - i_d(t)) + K_{di} \int_0^t i_d^*(\tau) - i_d(\tau) d\tau \quad (98)$$

$$u_q(t) = K_{dq} (i_q^*(t) - i_q(t)) + K_{qi} \int_0^t i_q^*(\tau) - i_q(\tau) d\tau \quad (99)$$

where K_{dp} and K_{di} represent the proportional and integral gains of the direct-current PI, respectively, whilst K_{qp} and K_{qi} are used for the quadrature current PI. The direct current setpoint i_d^* is set to zero to reduce the copper power loss, whereas i_q^* is obtained by substituting the MPC-calculated control force in (21). Current controllers operate at a sampling rate higher than that of the MPC to effectively utilize the first-order holds and provide smooth quadrature current trajectories. Finally, the converter control signals are sent to a sinusoidal pulse-width modulator to generate the switching states $S_{1,2,3}$ for the AC/DC converter.

V. RESULT AND DISCUSSION

A. WAVE EXCITATION FORCE ESTIMATION

A simulation study was conducted using MATLAB, and the performances of the SR-UKF and AR models were evaluated under realistic sea conditions. The point-absorber parameters are listed in Table 1. The SR-UKF operates at a sampling period of $T_s = 10$ ms and its tuning parameters are set as $\zeta = 0.9$, $\kappa = 1$, and $b = 2$. The process noise covariance matrix was selected as $q = \text{diag}([10^{-2} \ 10^{-2} \ 10^{-10} \ 10^{-10} \ 10^{-10} \ 10^{-10} \ 10^{-10} \ 10^{-10}])$, whereas the measurement noise was $r = 10^{-2}$.

Figure 3 shows the estimation performance of SR-UKF in the presence of regular monochromatic waves. The wave

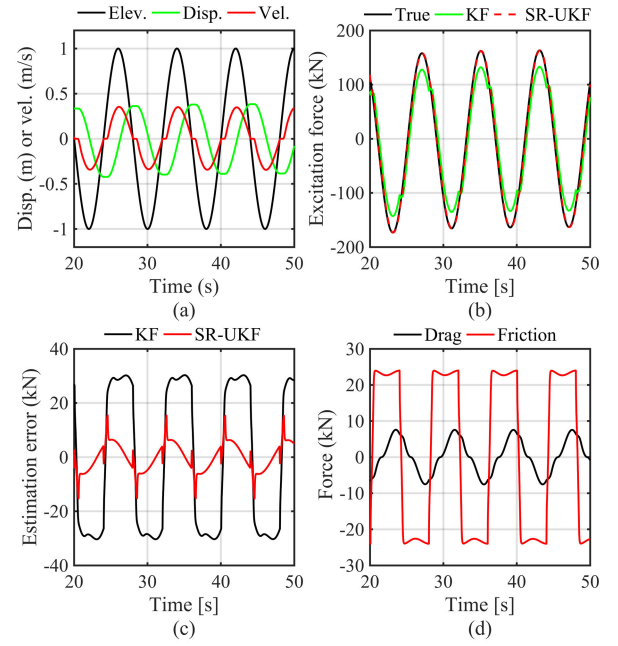


FIGURE 3. Wave excitation force estimation under a monochromatic wave of significant height H_s of 2 m and energy period T_e of 8 s: (a) wave elevation, buoy displacement, and buoy velocity, (b) excitation force estimations, (c) estimation errors, (d) drag force and friction force.

TABLE 2. Applied sea-states and their characteristics.

Parameter, symbol (unit)	S1	S2	S3	S4	S5
Significant height, H_s (m)	1.52	3.09	3.66	5.18	7.43
Energy period, T_e (s)	6.4	7.8	9.4	10.1	12.3

elevations, depicted in Fig. 3(a), cause a buoy heaving displacement that is fed as an input along with the control force $f_u(t)$. In Fig. 3(b), the estimation performance of the SR-UKF outperformed the linear KF, as it achieved a short-duration maximum estimation error of 14.34 kN when the excitation force changed direction. Meanwhile, the linear KF was less accurate as it gave a maximum error of 26.86 kN, because it is based on the linear point absorber model in (27) and (28) which does not acknowledge the drag and friction forces shown in Fig. 3(d). Appendix A provides a brief description of the linear Kalman design.

Realistic sea waves are not monochromatic, but polychromatic and irregular in amplitude, frequency, and phase. In this work, polychromatic waves were based on the Pierson–Moskowitz spectrum because they provide a better description of fully developed sea states [41]. A polychromatic wave is applied in Fig. 4 to assess the estimation performance in a more realistic environment. Unsurprisingly, the SR-UKF exceeded the linear KF in terms of the excitation force estimation, as shown in Fig. 4b. The SR-UKF is further subjected to five polychromatic sea-state conditions to evaluate its estimation performance against varying sea characteristics. Table 2 lists the characteristics of the sea states applied. Furthermore, buoy displacement